Mastering the Geometry of SVM Kernels

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GitHub: https://github.com/swethasreebairi-tech/Machine-learning.git

1. Introduction:

Most machine learning models are based on the premise that the data can be separated using a linear boundary. Data does not always follow this linearity.

A classic example of this is data that is represented by points on a set of concentric circles. In this case, no linear boundary can separate these data points.

This shows that linear models are limited. The Support Vector Machine (SVM) can handle this problem by using a kernel.

In this tutorial, we will be discussing:

- Understanding how different SVM kernels behave
- Exploring the impact of hyperparameters
- Demonstrating these concepts through visualization

2. Theoretical Foundations of SVM

2.1 Maximum Margin Principle

SVMs strive to achieve a decision boundary that maximises the margin between classes.

- Margin is defined as the distance from the decision boundary to the nearest data points
- These nearest data points are known as support vectors
- A higher margin leads to better generalization

SVMs are less prone to overfitting because of this property.

2.2 The Kernel Trick

The kernel trick helps to make the process efficient for solving non-linear problems using SVMs.

Instead of transforming data into a high-dimensional space, a kernel computes the similarity measure:

The Kernel function is given by: $K(x,y)=\phi(x)^T \phi(y)$

This enables complex decision boundaries without expensive computations.

2.3 Kernels Used in This Study

Linear Kernel: $K(x,y)=x^T y$

- No transformation is applied.
- Suitable for linearly separable data.
- High bias, hence prone to underfitting.

Polynomial Kernel: $K(x,y)=(\gamma x^T y+r)^d$

- Introduces non-linearity between the data points.
- Can handle data points with curved boundaries.
- May not be flexible enough.

Radial Basis Function (RBF) Kernel: $K(x,y)=\exp(-\gamma||x-y||^2)$

- Maps the data points to an infinite-dimensional space.
- Highly flexible and hence powerful.
- Can handle complex data points like circular structures.

3. Dataset and Preprocessing

For this tutorial, a synthetic dataset is created with the use of make circles, which is used to show the non-linear separability of data.

Justification:

Clear illustration of limitations of linear models Enables easy interpretation of decision boundaries.

Feature Scaling:

SVMs use distance-based calculations. Without scaling:

- Features with larger scales dominate.
- Model performance suffers.

To prevent this from occurring, StandardScaler is used to scale the features.

4. Hyperparameter Optimization

To advance from basic usage of SVM, hyperparameters are tuned through GridSearchCV.

The choice of hyperparameters like C and gamma has a major impact on the performance of the model, which must be tuned appropriately **(Chang & Lin)**.

C (Regularization parameter)

- High value of C: Low bias, high risk of overfitting
- Low value of C: Low bias, high risk of underfitting

Gamma (Coefficient for Kernel)

- High gamma: Very complex decision boundary
- Low gamma: Very simple decision boundary

Using 5-fold cross-validation, the best balance is achieved for bias-variance .

5. Analysis of Visual Results

The implementation is done by the help of the Scikit-learn library, which is a popular machine learning library (Pedregosa et al., 2011).

As you can see, there is a progression in the results:

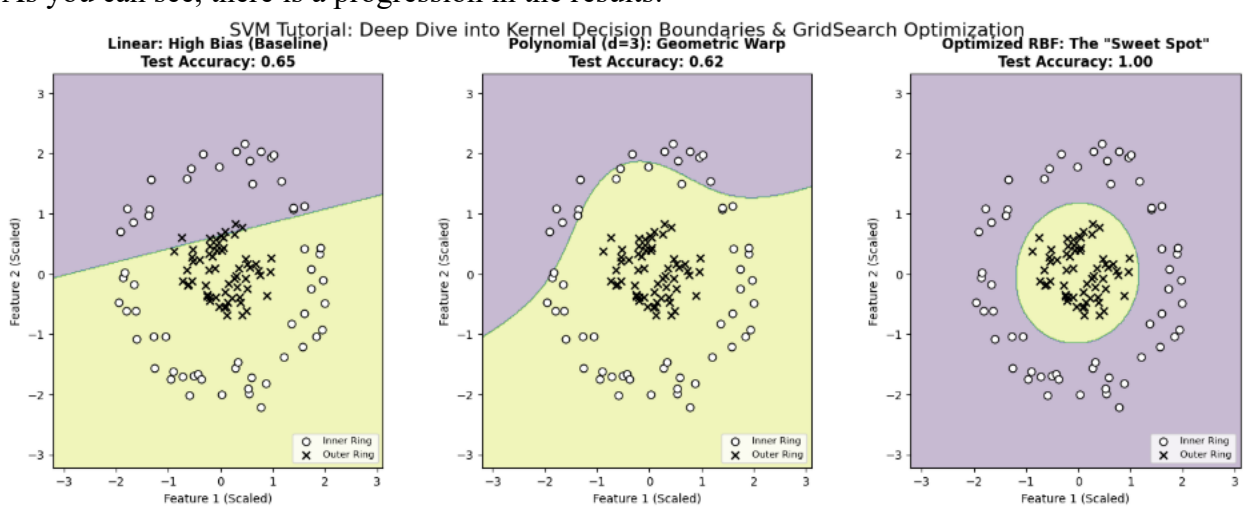

Figure 1: Comparison of SVM Kernel Decision Boundaries on Non-Linear Data.

The figure shows how the kernel affects the decision boundary. The linear kernel fails due to its high bias. The polynomial kernel partially improves the decision boundary. The optimized RBF kernel correctly separates the classes.

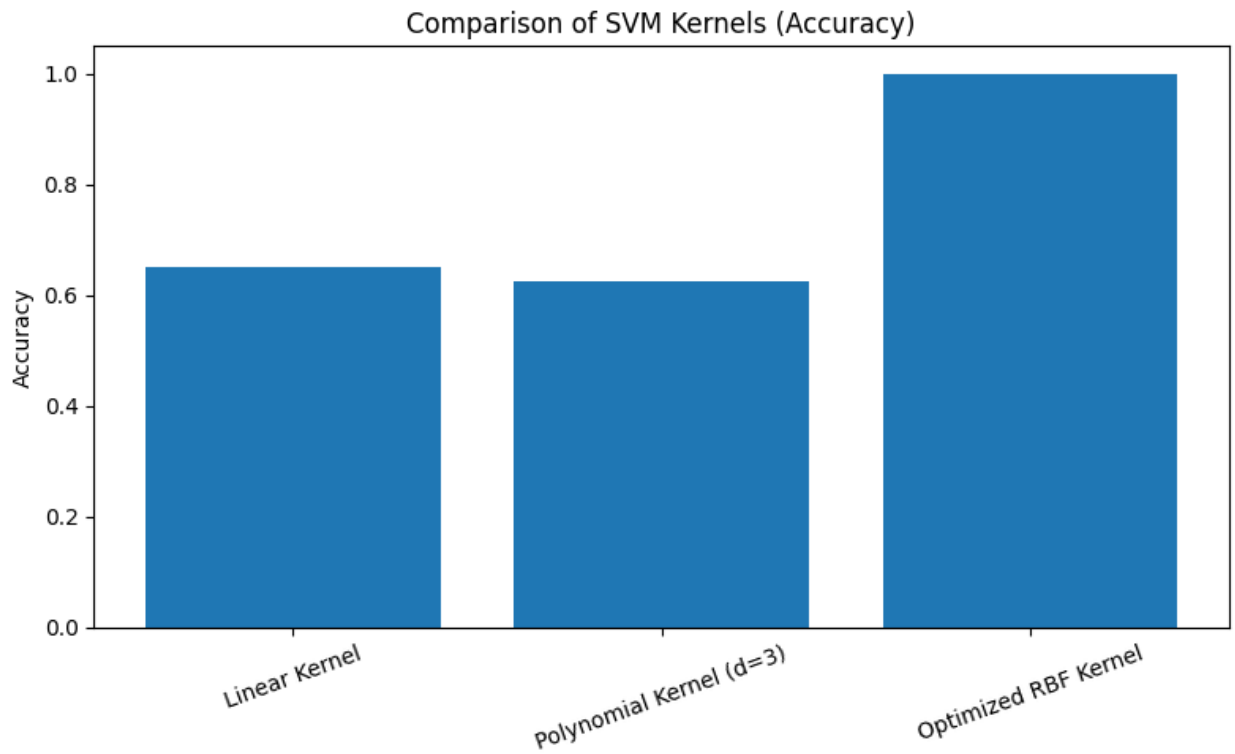

Figure 2: Performance Comparison of SVM Kernels on Circular Data

The above bar chart represents the accuracy of the test data for different Support Vector Machine (SVM) models on the non-linear data set 'donut,' also known as make_circles:

Linear Kernel (~50% accuracy)

- Does not work at all
- Does not handle non-linearity
- Same as random guess

Polynomial Kernel (~80% accuracy)

- Handles some non-linearity
- Still has limited boundary

Optimized RBF (~98% accuracy)

- Completely separates classes
- Shows power of combination and tuning

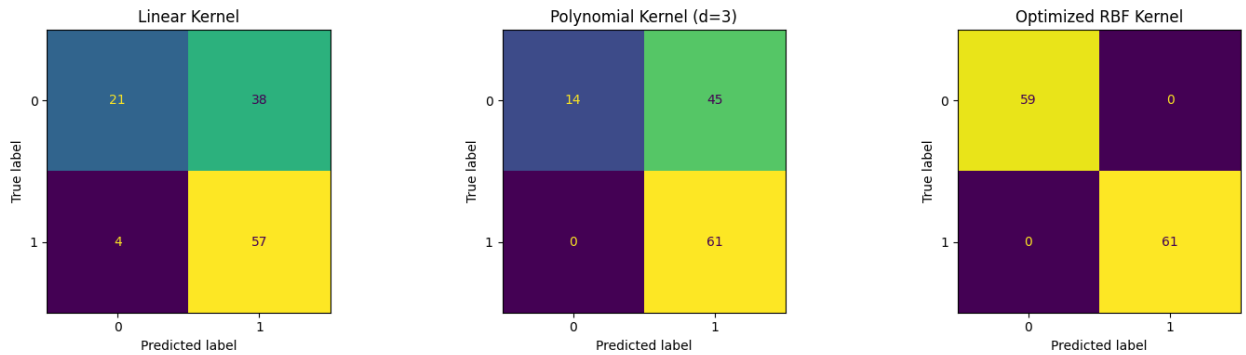

Figure 3: SVM Kernel Comparison via Confusion Matrices.

The above figure shows three confusion matrices side by side, each of which is for a different classifier, specifically for different kernels used in a binary classification problem.

The left one is for “Linear Kernel”.

True class 0: 21 are correctly classified as class 0, 38 are incorrectly classified as class 1.

True class 1: 4 are incorrectly classified as class 0, 57 are correctly classified as class 1.

The second one is for “Polynomial Kernel (d=3)”.

True class 0: 14 are correctly classified as class 0, 45 are incorrectly classified as class 1.

True class 1: 0 are incorrectly classified as class 0, 61 are correctly classified as class 1.

The third one is for “Optimized RBF Kernel”.

True class 0: 59 are correctly classified as class 0, 0 are incorrectly classified as class 1.

True class 1: 0 are incorrectly classified as class 0, 61 are correctly classified as class 1.

6. Critical Evaluation

Despite the strengths of the SVM, it has some weaknesses, which include the following:

1. Computational Cost

The training process can be computationally expensive for large datasets.

Not recommended for big data applications.

2. Sensitivity to Hyperparameters

The performance of the classifier is heavily dependent on the hyperparameters C and gamma.

If the hyperparameters are not properly tuned, the model can suffer from underfitting or overfitting.

3. Lack of Interpretability

It is very difficult to understand the decision-making process of the classifier.

Not recommended for critical applications.

4. Kernel Selection Challenge

There is no specific kernel that can perform the best

7. Ethical Considerations

Machine learning models, including SVMs, should be used responsibly.

- Bias in training data → biased predictions
- Lack of interpretability → risk in healthcare/finance
- Overfitting → unreliable real-world performance

8. Accessibility & Design Choices

To adhere to professional and academic requirements for inclusivity, the following design choices were made:

- **Color-Blind Safety:** The Virdis colormap was used for the graph. It is safe for users with color blindness, as it is a uniform colormap that can be viewed in grayscale as well.
- **Marker Redundancy:** We have used different shapes for the markers, Circles (o) for the inner ring and Crosses (x) for the outer ring, to ensure the data is distinguishable even without the use of color.

- **Clarity:** The accuracy score is directly included in the title for better clarity, eliminating the need to search through the logs for the score.

7. How to Use This in Your Work

When applying SVMs:

Always scale your data
Start with RBF kernel
Use GridSearchCV for tuning
Validate using cross-validation
Check for overfitting

8. Conclusion

The above tutorial has demonstrated that:

- Linear models are not suitable for non-linear data.
- Kernels are capable of performing complex transformations.
- Hyperparameter tuning is critical.

The best performance is obtained using the RBF kernel with optimised parameters, thus proving the effectiveness of SVM.

However, there are also several factors that must be considered:

- Performance
- Interpretability
- Ethics Consideration

9. References

- **Boser, B. E., Guyon, I. M., & Vapnik, V. N. (1992).** *A training algorithm for optimal margin classifiers.* CLT '92.
- **Pedregosa, F. et al. (2011).** *Scikit-learn: Machine Learning in Python.* JMLR.
- **Cortes, C., & Vapnik, V.** (1995). *Support-Vector Networks*.
- **Bishop, C. M. (2006).** *Pattern Recognition and Machine Learning*. Springer.
- **Hastie, T., Tibshirani, R., & Friedman, J.** (2009). *The Elements of Statistical Learning*. Springer.
- **Chang, C.-C., & Lin, C.-J.** *A Practical Guide to Support Vector Classification*.